Hyungu Kim

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Education

Yonsei University Mar. 2020 – Expected Feb. 2027
B.S. in Electrical and Electronic Engineering *Seoul, South Korea*
GPA: 3.93 / 4.30 (Cumulative), 4.14 / 4.30 (Major)

Awards and Honors

- **1st Place, Phase 1 and Phase 2**, Berkeley AgentX–AgentBeats Competition (2026).
- **Academic Excellence Award**, Yonsei University (Spring 2021, Spring 2025).
- **Next-Generation Engineer Badge**, IPESK.

Research Interests

- **Physics-aware agents**: physical reasoning for generation, manipulation, and navigation.
- **Knowledge accumulation**: persistent memory, knowledge sharing, and retrieval across agents.
- **Architecture exploration**: model probing, controlled ablations, and scalable learning methods.

Research Experience

SNU Visual & Geometric Intelligence (VGI) Lab **Research Intern** Oct. 2025 – Present
Seoul, South Korea

- Built a **System 1.5 VLA harness** for Berkeley AgentBeats, contributing to first-place finishes in both phases.
- Pretrained and skill-tuned VLA policies for **BEHAVIOR-1K**; developed VLM-based belief-graph management (self-reported Q-score: **0.45**).
- Developed a **Recursive Language Model** system for quantitative physics reasoning from visual inputs.

Yonsei Multimedia AI Lab **Research Intern** Dec. 2025 – Jun. 2026
Seoul, South Korea

- Researched **CAD-free 6-DoF pose tracking** with Hunyuan3D-generated object meshes.
- Probed **MASt3R** dense features for granularity-aware matching across large viewpoint changes.

SNU PI Lab **Research Intern** Mar. 2025 – Jun. 2025
Seoul, South Korea

- Studied **sim-to-real robot navigation** with vision-language-action models.
- Developed navigation waypoint generation from combined **natural-language and sketch instructions**.

Projects

- **Efficient ResNet-34 Inference.** Applied structural re-parameterization, knowledge distillation, and quantization; reduced MACs by **92%** and accelerated inference by **7**×.
- **Video-Based Vehicle Speed Estimation.** Built a BiLSTM model to predict ego-vehicle speed from six consecutive KITTI frames.
- **Deep Learning Reproductions.** Implemented **21 research papers**, including Grad-CAM and XOR-Net, with experimental reports.
- **Reinforcement Learning Studies.** Investigated Q-learning, SAC, offline RL, model-based RL, and RLHF through technical reports.

Technical Skills

- **Programming Languages:** Python, C/C++, Java, JavaScript, TypeScript.
- **Machine Learning & Robotics:** PyTorch, JAX, Hugging Face Transformers, vLLM, NVIDIA Isaac Sim.
- **Software & 3D Development:** Node.js, Three.js, FastAPI, Blender (bpy).
- **Developer Tools & Infrastructure:** Git, Docker, Linux, CUDA.